

MTLV HR PORTAL

Revised Project Requirement Document – Version 2.0

Incorporating: Admin Role | All Document Types | Salary Increment Notifications | Full Form Field Specifications | System Flow

Document Info	
Prepared By	MTLV Development Team
Version	2.0 (Revised & Extended)
Previous Version	1.0 – Base PRD
Status	Draft – Awaiting Sign-off
Date	June 2025

1. What Changed from Version 1.0

Version 2.0 extends the original PRD with the following additions and refinements:

Change Area	Description
Admin Role Added	A new Super Admin layer sits above HR with full access across all modules and approval authority.
HR Role Restricted	HR can perform day-to-day operations but cannot access Admin-only settings, approve salary increments, or delete records.
New Document Types	Bond / Notice Period Letter added. All document types enumerated per module.
Salary Increment Notification	Automated 15-day advance notification to HR and Admin before each employee's 1-year anniversary (increment trigger).
Student / Intern Module Expanded	Certificate covers internship, training, and project students. All certificate fields specified.
Full Form Field Specs	Every HR-facing form now has a complete field-level specification table (label, type, required, validation).
System Flow	End-to-end flow documented per module with Actor, Action, System Response, and Output columns.
Re-joining / Ex-Employee Flow	New sub-module: HR can re-onboard ex-employees and generate experience / relieving certificates for them.

2. Roles & Access Control

2.1 Role Overview

The system now supports two roles only: Admin and HR. No Employee self-service portal exists.

Role	Description
Admin (Super Admin)	Full system access. Can manage HR user accounts, approve salary increments, configure notification rules, view all audit logs, and perform any action HR can perform.
HR	Operational access. Can onboard employees, manage exit workflows, generate documents, manage salary slips, and handle student certificates. Cannot access Admin-only settings.

2.2 Feature Access Matrix

Feature / Module	Admin	HR
Add / Edit Employee	Full Access	Full Access
View Employee List	Full Access	Full Access
Delete Employee Record	Yes (soft delete)	No
Upload / View Documents	Full Access	Full Access
Initiate Exit Process	Full Access	Full Access
Approve Exit Completion	Yes	Can complete; Admin reviews
Generate Relieving Letter	Full Access	Full Access
Generate Experience Letter	Full Access	Full Access
Generate Bond / Notice Period Letter	Full Access (Edit also)	Full Access
Define Salary Structure	Full Access	Full Access
Generate Monthly Salary Slips	Full Access	Full Access
Approve Salary Increment	Yes (Admin Only)	Initiates request only
Salary Increment Notification Config	Yes (Admin Only)	View only
Add / Edit Student	Full Access	Full Access
Generate Student Certificate	Full Access	Full Access
Bulk PDF Download	Full Access	Full Access
Import / Export Excel	Full Access	Full Access
Manage HR User Accounts	Yes (Admin Only)	No

View Audit Logs	Full Access	Own actions only
Dashboard	Full (all stats)	Standard stats
System Settings / Notifications Config	Yes (Admin Only)	No
Re-onboard Ex-Employee	Full Access	Full Access

3. Document Types – Complete Master List

All documents in the system are PDF-generated from HTML/CSS templates using WeasyPrint. Below is the complete catalogue.

Document Name	Module	Generated By	Auto / Manual
Offer Letter	Onboarding	Admin / HR	Manual (upload or generate)
Appointment Letter	Onboarding	Admin / HR	Manual
Bond / Service Agreement Letter	Onboarding	Admin / HR	Manual – triggered by bond period set
Notice Period Letter	Onboarding / Exit	Admin / HR	Manual
Salary Increment Letter	Salary / Increment	Admin	Auto-generated after approval
Salary Slip (Payslip)	Salary Slip	HR / Admin	Monthly – manual trigger or bulk
Relieving Letter	Exit	HR / Admin	After exit completion
Experience Letter	Exit	HR / Admin	After exit completion
No-Objection Certificate (NOC)	Exit (optional)	Admin / HR	On request
Full & Final Settlement Letter	Exit (optional)	Admin / HR	On request
Internship Certificate	Student – Intern	HR / Admin	After completion date if her want to generate the certificate after the completion period like 6 month duration then warning is shown to HR if she say yes then the certificate is generated and the notification is shown to the admin
Training Completion Certificate	Student – Training	HR / Admin	After completion date
Project Completion Certificate	Student – Project	HR / Admin	After completion date

4. Salary Increment Notification System

4.1 Business Rule

Every employee is eligible for a salary review after completing 1 year of service. The system must automatically calculate each employee's 1-year anniversary date and send advance notifications to both HR and Admin 15 days before that date, giving them time to review and approve a salary increment.

4.2 Trigger Logic

Parameter	Value / Rule
Trigger Event	Employee completes (Joining Date + 365 days) - 15 days = Notification Date
Notification Recipients	All Admin users + All HR users (email + in-app notification)
Notification Lead Time	15 calendar days before the 1-year anniversary
Notification Repeat	Day -15, Day -7, and Day 0 (anniversary day) if still unactioned
Scheduler	Django-Q2 or Celery Beat – runs daily at 08:00 to check upcoming anniversaries
Action Required	Admin reviews the notification, opens Salary Increment form, approves/modifies, and saves
Document Generated	Salary Increment Letter (PDF) auto-generated upon Admin approval
Employee Notification	Employee receives increment letter via email after Admin approval

4.3 Notification Flow

1. Celery Beat task runs daily at 08:00.
2. Query: SELECT all active employees WHERE (joining_date + 350 days) <= today (i.e., anniversary in 15 days).
3. For each matched employee: create a SalaryIncrementReminder record (if not already created for this cycle).
4. Send email notification to all Admin users and HR users.
5. Create in-app notification visible on Admin and HR dashboards.
6. Admin opens the Salary Increment Approval form, enters new salary details, and clicks Approve.
7. System saves new salary structure effective from anniversary date.
8. System generates Salary Increment Letter PDF and emails it to the employee.
9. Increment reminder is marked Actioned – no further reminders sent for this cycle.

4.4 Salary Increment Approval Form – Field Specification

Field / Label	Input Type	Required	Notes / Validation
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Employee Name	Read-only text	Auto	Populated from employee record
Employee ID	Read-only text	Auto	Auto-populated
Department	Read-only text	Auto	Auto-populated
Joining Date	Read-only date	Auto	Auto-populated
1-Year Anniversary Date	Read-only date	Auto	Calculated: joining_date + 365
Current Basic Salary	Read-only currency	Auto	From active salary structure
Current Net Salary	Read-only currency	Auto	Calculated from structure
New Basic Salary	Number input	Yes	Must be >= current basic salary
New HRA	Number input	Yes	Can match or exceed current HRA
New Allowances	Number input	Yes	Optional increase
Increment Amount	Read-only currency	Auto	New Net - Current Net (calculated)
Increment Percentage	Read-only %	Auto	$((\text{New}-\text{Old})/\text{Old}) \times 100$
Effective Date	Date picker	Yes	Defaults to anniversary date; Admin can adjust
Increment Reason	Textarea	Yes	Min 20 characters; appears in letter
Approved By	Read-only (logged-in Admin)	Auto	Current Admin user
Approval Date	Read-only	Auto	Today's date on submission
Notes / Remarks	Textarea	No	Internal notes; not in letter

5. Module 1 – Employee Onboarding (Revised)

5.1 Add New Employee Form – Field Specification

Field / Label	Input Type	Required	Notes / Validation
First Name	Text input	Yes	Max 100 chars; alpha + spaces only
Last Name	Text input	Yes	Max 100 chars
Email Address	Email input	Yes	Must be unique across all employees
Phone Number	Tel input	Yes	10-digit Indian mobile; format: 9XXXXXXXXXX
Alternate Phone	Tel input	No	Optional secondary contact
Date of Birth	Date picker	Yes	Employee must be >= 18 years old

Gender	Dropdown	Yes	Male / Female / Other / Prefer not to say
Address Line 1	Text input	Yes	Max 200 chars
Address Line 2	Text input	No	Optional
City	Text input	Yes	Max 100 chars
State	Dropdown	Yes	Indian state list
PIN Code	Text input	Yes	6-digit numeric
Department	Dropdown	Yes	Populated from Department master table
Designation	Text input	Yes	Max 100 chars
Employment Type	Dropdown	Yes	Full-time / Part-time / Contract / Intern
Joining Date	Date picker	Yes	Cannot be future date > 30 days out
Notice Period (days)	Number input	Yes	Default 30; used in notice period letter
Bond Period (months)	Number input	No	If > 0, triggers Bond Letter generation
Emergency Contact Name	Text input	Yes	Max 100 chars
Emergency Relationship	Dropdown	Yes	Spouse / Parent / Sibling / Friend / Other
Emergency Phone	Tel input	Yes	10-digit mobile
Employee ID	Read-only text	Auto	Auto-generated: EMP-YYYY-XXXX
Aadhaar Number	Text input (masked)	No	12-digit; stored encrypted
PAN Number	Text input (masked)	No	10-char alphanumeric; stored encrypted
Profile Photo	File upload	No	JPEG/PNG; max 2 MB

5.2 Document Upload Form – Field Specification

Field / Label	Input Type	Required	Notes / Validation
Employee	Read-only / pre-selected	Auto	Linked to employee context
Document Type	Dropdown	Yes	Resume / Aadhaar / PAN / Offer Letter / Appointment Letter / Bond Letter / Other
Document Label	Text input	Conditional	Required if Document Type = Other; max 80 chars
File	File upload	Yes	PDF/JPEG/PNG; max 10 MB per file
Upload Date	Read-only	Auto	System date on upload
Remarks	Textarea	No	Optional notes about this document

5.3 Onboarding Flow

Step	Actor	Action	System Response
1	HR / Admin	Opens Add Employee form	Blank form rendered with auto-ID preview
2	HR / Admin	Fills all required fields, clicks Save	Validates all fields; creates Employee record; auto-assigns EMP-ID
3	System	Post-save check: Bond Period > 0?	If yes, creates a draft Bond Letter record; notifies HR to review & generate
4	HR / Admin	Uploads documents (Aadhaar, PAN, Offer Letter, etc.)	Files stored in /media/employees/{emp_id}/; linked to employee
5	System	Logs activity in Audit Log	Activity Feed updated: 'New employee added – [Name]'
6	HR / Admin	Opens Generate Letters menu	Options: Offer Letter, Appointment Letter, Bond Letter (if applicable)
7	HR / Admin	Clicks Generate Offer/Appointment Letter	WeasyPrint renders PDF from template; stored and available for download

6. Module 2 – Employee Exit Management (Revised)

6.1 Initiate Exit Form – Field Specification

Field / Label	Input Type	Required	Notes / Validation
Employee	Searchable dropdown	Yes	Only active employees; search by name or EMP-ID
Resignation Date	Date picker	Yes	Date HR received resignation letter
Last Working Day	Date picker	Yes	Resignation Date + Notice Period days (auto-calculated; editable)
Exit Type	Dropdown	Yes	Resignation / Termination / Retirement / Contract End / Absconding
Exit Reason (HR notes)	Textarea	Yes	Internal note; min 20 chars; not shared with employee
Notice Period Waiver	Checkbox	No	If checked: Last Working Day may differ from notice period end
Notice Period Letter Required	Checkbox	No	If checked: system generates Notice Period Letter PDF
HR Remarks	Textarea	No	Additional internal remarks
Initiated By	Read-only (logged-in user)	Auto	Current HR / Admin user
Initiated Date	Read-only	Auto	Today's date

6.2 Employee Exit Self-Service Form (Secure Link) – Fields

Sent to employee via secure, time-bound link. No login required. Employee fills and submits.

Field / Label	Input Type	Required	Notes / Validation
Employee Name	Read-only	Auto	Pre-filled from record
Employee ID	Read-only	Auto	Pre-filled
Last Working Day	Read-only	Auto	Pre-filled
Asset Return – Laptop / Desktop	Checkbox + text	Yes	Must confirm returned or state reason
Asset Return – ID Card	Checkbox + text	Yes	
Asset Return – Access Card / Keys	Checkbox + text	Yes	
Asset Return – Other Assets	Textarea	No	List any other company assets
Knowledge Transfer Status	Dropdown	Yes	Completed / In Progress / Not Applicable
KT Handover To (Name)	Text input	Conditional	Required if KT Status = Completed / In Progress
KT Remarks	Textarea	No	Details of handover
Reason for Leaving (Employee)	Dropdown	Yes	Better Opportunity / Higher Studies / Personal / Relocation / Other
Reason Details	Textarea	Conditional	Required if Reason = Other; min 20 chars
Work Environment Rating	Radio (1-5)	Yes	1 = Very Poor, 5 = Excellent
Management Rating	Radio (1-5)	Yes	Used to send relieving/experience letter after exit
Would Recommend Company	Radio (Yes/No/Maybe)	Yes	
Personal Email (for future)	Email input	Yes	
Personal Phone (for future)	Tel input	No	Optional contact for alumni
Any Feedback / Suggestions	Textarea	No	Open feedback; max 1000 chars
Digital Signature / Declaration	Checkbox	Yes	"I confirm all information is accurate."

6.3 Documents Generated at Exit

Document	Trigger Condition
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Relieving Letter	Exit status = Completed; generated by HR / Admin
Experience Letter	Exit status = Completed; generated by HR / Admin
Notice Period Letter	If 'Notice Period Letter Required' was checked at initiation
No-Objection Certificate (NOC)	Optional; generated on HR / Admin request
Full & Final Settlement Letter	Optional; Admin generates after F&F calculation

6.4 Exit Flow

Step	Actor	Action	System Response
1	HR / Admin	Opens Initiate Exit form, selects employee, fills fields	Creates ExitRequest record; status = Pending
2	System	Sends secure email to employee's work email	Email contains unique UUID-based link; expires in 7 days
3	Employee	Opens link; fills asset return, KT, exit questionnaire	Status -> In Progress on first field fill; partial save supported
4	Employee	Clicks Submit with digital declaration checked	Status -> Completed; HR and Admin notified via email + in-app
5	HR / Admin	Reviews exit form responses	All fields visible in read-only exit detail view
6	HR / Admin	Clicks Generate Relieving Letter	WeasyPrint PDF rendered; stored; download available
7	HR / Admin	Clicks Generate Experience Letter	Same as above; separate template
8	Admin	Optionally generates NOC / F&F Letter	Admin-only action; PDF stored and emailed if required
9	System	Employee record marked as Exited (soft delete)	Active employee count decremented on dashboard

7. Module 2B – Re-joining / Ex-Employee Flow

7.1 Overview

When an ex-employee rejoins the company, HR can re-activate their record or create a new linked record. The system maintains the full history of the previous tenure.

7.2 Re-joining Form – Field Specification

Field / Label	Input Type	Required	Notes / Validation
Search Ex-Employee	Searchable dropdown	Yes	Search by name, email, or previous EMP-ID

Previous EMP-ID	Read-only	Auto	Displayed after selection
Previous Last Working Day	Read-only	Auto	From exit record
New Joining Date	Date picker	Yes	Start date of new tenure
New Designation	Text input	Yes	May differ from previous designation
New Department	Dropdown	Yes	May differ from previous department
New Employment Type	Dropdown	Yes	Full-time / Contract / etc.
New Employee ID	Read-only	Auto	New EMP-ID auto-generated for new tenure
Salary Structure	Form section	Yes	Fresh salary structure to be filled
HR Remarks	Textarea	No	Reason for re-joining, notes

7.3 Ex-Employee Certificates

HR can generate certificates for ex-employees at any time, even after exit. Available documents:

- Relieving Letter (backdated to original Last Working Day)
- Experience Letter (covering total tenure duration)
- Service Certificate
- Any previously generated document can be re-downloaded from the employee's historical record

8. Module 3 – Salary Slip Management (Revised)

8.1 Salary Structure Form – Field Specification

Field / Label	Input Type	Required	Notes / Validation
Employee	Searchable dropdown	Yes	Active employees only
Effective From Date	Date picker	Yes	Date from which this structure applies
Basic Salary	Currency input	Yes	Gross base; numeric; min Rs. 1
HRA (House Rent Allowance)	Currency input	Yes	Can be 0
Conveyance Allowance	Currency input	No	Default 0
Medical Allowance	Currency input	No	Default 0
Special Allowance	Currency input	No	Default 0
Other Allowances	Repeatable row (label + amount)	No	HR can add custom allowance lines
Monthly Bonus	Currency input	No	Recurring monthly bonus; default 0
Provident Fund (PF)	Currency input /	No	Fixed amount or % of basic

	%		
Professional Tax	Currency input	No	State-specific; default 200
TDS (Income Tax Deduction)	Currency input	No	Monthly TDS deduction
Other Deductions	Repeatable row (label + amount)	No	HR can add custom deduction lines
Gross Salary	Read-only currency	Auto	Sum of all earnings
Total Deductions	Read-only currency	Auto	Sum of all deductions
Net Salary	Read-only currency	Auto	Gross - Total Deductions
Remarks	Textarea	No	Internal notes

8.2 Generate Salary Slip Form – Field Specification

Field / Label	Input Type	Required	Notes / Validation
Month	Dropdown (Jan-Dec)	Yes	Select payroll month
Year	Number / Dropdown	Yes	4-digit year; default current year
Employee Selection	Radio	Yes	All Active Employees / Specific Employee
Specific Employee	Searchable dropdown	Conditional	Required if selection = Specific Employee
Working Days in Month	Number input	Yes	Default = total calendar working days; editable
Actual Days Worked	Number input	Conditional	Required if employee was on LOP
Loss of Pay (LOP) Days	Number input	No	Auto-calculated from (Working Days - Actual Days); salary proportionally reduced
One-time Bonus (this month)	Currency input	No	Added to this slip only; does not affect structure
One-time Deduction (this month)	Currency input	No	Added to this slip only
Payslip Number	Read-only	Auto	Format: PS-YYYY-MM-XXXX
Payment Date	Date picker	Yes	Date salary was / will be credited
Payment Mode	Dropdown	No	Bank Transfer / Cheque / Cash

9. Module 4 – Student Certificate Management (Revised)

9.1 Add Student / Intern Form – Field Specification

Field / Label	Input Type	Required	Notes / Validation
Student Name	Text input	Yes	Max 100 chars
Email Address	Email input	Yes	Unique per training program
Phone Number	Tel input	No	Optional
Date of Birth	Date picker	No	Optional
Institute / College Name	Text input	Yes	Max 200 chars
Course / Program (at institute)	Text input	Yes	e.g., B.Tech CSE, MBA, BCA
Student Type	Dropdown	Yes	Intern / Trainee / Project Student / Industrial Visit
Training Program Name (internal)	Text input	Yes	e.g., Summer Internship 2025, Python Training Batch 3
Department (at MTLV)	Dropdown	Yes	Which internal department they worked in
Mentor / Supervisor Name	Text input	No	MTLV employee who supervised them
Joining Date	Date picker	Yes	First day at MTLV
Completion Date	Date picker	Yes	Last day; can be future (generates certificate after this date)
Training Duration	Read-only	Auto	Auto-calculated: days/weeks/months between joining and completion
Project / Work Description	Textarea	No	Brief description; optionally appears in certificate
Certificate Type	Dropdown	Yes	Internship Certificate / Training Certificate / Project Completion Certificate
Certificate Number	Read-only	Auto	Format: CERT-YYYY-XXXXXX
Status	Dropdown	Yes	Active / Completed / Discontinued
Remarks	Textarea	No	Internal notes

9.2 Certificate Generation Flow

Step	Actor	Action	System Response
1	HR / Admin	Opens Student list; clicks student	Student detail page opens

2	HR / Admin	name Clicks Generate Certificate	System checks: Completion Date <= today? If no, warns but allows override with Admin permission
3	System	Renders certificate PDF via WeasyPrint template	Fills: Student Name, Course, Institute, Duration, CERT number, Company logo, signature
4	System	Stores PDF in /media/certificates/{cert_number}/	Certificate marked as Generated in database
5	HR / Admin	Downloads individual PDF or selects multiple for ZIP	ZIP generated with all selected certificates
6	HR / Admin	Optionally emails certificate to student	System sends email with PDF attachment to student's email address

10. Notification System

10.1 Notification Types

Notification Type	Trigger	Recipients	Channel
Salary Increment Reminder (Day -15)	15 days before 1-year anniversary	All Admin + All HR	Email + In-app
Salary Increment Reminder (Day -7)	7 days before 1-year anniversary (if unactioned)	All Admin + All HR	Email + In-app
Salary Increment Reminder (Day 0)	Anniversary day (if still unactioned)	All Admin + All HR	Email + In-app
Increment Approved	Admin approves salary increment	HR + Employee	Email + In-app
Exit Form Submitted	Employee submits exit self-service form	HR + Admin	Email + In-app
Exit Initiated	HR initiates exit process	Admin	In-app
Secure Link About to Expire	6 days after sending exit link (1 day before 7-day expiry)	HR + Admin	Email + In-app
New Employee Added	HR saves new employee record	Admin	In-app
Salary Slips Generated (Bulk)	Bulk generation job completed	HR who triggered it	In-app
Document Expiry Warning	Aadhaar / contract expires within 30 days (future feature)	HR + Admin	Email + In-app

Certificate Generated	HR generates a student certificate	Admin (summary)	In-app
Student fess	HR take fess amount in the installment also show the what is the fees of the student and in which installment they pay if student is not paying the installment at time then the a soct and professional email sent to the students that your installment is not pay please contact to the HR	Admin have notification all installment	

10.2 In-App Notification Bell

- All notifications appear in a notification bell icon in the top navigation bar.
- Unread count badge shown on the bell.
- Clicking the bell opens a slide-over panel with all recent notifications.
- Each notification has: icon (type), message, timestamp, and a link to the relevant page.
- Admin sees all system notifications. HR sees notifications relevant to their actions.
- Notifications can be marked as read individually or all at once.

11. Dashboard (Revised)

11.1 Admin Dashboard

Admin sees an extended dashboard with all statistics plus system health indicators.

Card / Widget	Data Shown
Total Employees	All employees (active + exited)
Active Employees	Employees with status = Active
Employees Exited This Month	Exit completions in current calendar month
Pending Exit Requests	Exit status = Pending or In Progress
Upcoming Salary Increments (15 days)	Count of employees whose anniversary is within 15 days
Salary Slips This Month	Count of slips generated for current month
Total Students (Active)	Student records with status = Active
Certificates Generated (This Month)	Certificates generated in current calendar month

HR Users Active	Count of active HR user accounts
Recent Activities Feed	Last 20 system-wide actions with actor, action, and timestamp
Pending Actions Widget	List of actionable items: unactioned increment reminders, pending exits, link expirations

11.2 HR Dashboard

HR sees operational metrics without Admin-only sections.

- Active Employees count
- My Recent Actions (actions performed by this HR user)
- Pending Exit Requests (initiated by this HR user)
- Salary Slips Generated This Month
- Students – Active count
- Certificates Generated This Month
- Upcoming Anniversaries (informational view – action is Admin-only)

12. Full System Flow – End to End

12.1 New Employee Lifecycle

Stage	Who Acts	What Happens
1. Onboarding	HR / Admin	Add employee -> upload documents -> generate offer/appointment letter -> set salary structure -> Bond letter if applicable
2. Active Employment	System (automated)	Daily check runs -> at Day 350 from joining, increment reminder fires to HR + Admin
3. Salary Increment Review	Admin	Admin receives notification -> opens Increment form -> reviews -> approves -> letter generated -> employee notified
4. Monthly Payroll	HR / Admin	Monthly: HR opens Generate Salary Slips -> selects month -> all active employees -> bulk PDF generated -> downloadable as ZIP
5. Exit Initiation	HR	HR starts exit for resigning employee -> sets last working day -> sends secure link email to employee
6. Employee Exit Form	Employee	Employee fills asset return, KT, questionnaire -> submits -> HR + Admin notified
7. Post-Exit Documents	HR / Admin	Generate Relieving Letter + Experience Letter + (optional) NOC, F&F Letter -> download/email to employee
8. Record Archival	System	Employee marked Exited (soft delete) -> all records retained -> searchable in exit records
9. Re-joining (if applicable)	HR	Search ex-employee -> re-onboard with new tenure details -> new EMP-ID issued -> new salary structure

12.2 Student / Intern Lifecycle

Stage	Who Acts	What Happens
1. Registration	HR / Admin	Add student with institute, course, joining date, completion date, certificate type
2. Active Training	HR	Student appears in active student list; status = Active
3. Completion	HR / Admin	On/after completion date: click Generate Certificate -> PDF rendered with CERT number
4. Certificate Delivery	HR	Download PDF -> optionally email to student directly from system
5. Bulk Operations	HR / Admin	Select multiple students -> Generate All -> Download ZIP
6. Export	HR / Admin	Export all student data to Excel for records

12.3 Salary Increment Automated Flow

Day	Event	System Action
Day 350 (Joining + 350)	15 days before anniversary	Celery Beat detects -> sends email + in-app to HR + Admin -> creates SalaryIncrementReminder
Day 358 (Joining + 358)	7 days before anniversary (if unactioned)	Second reminder email + in-app
Day 365 (Joining + 365)	Anniversary day (if still unactioned)	Final reminder: 'Action Required Today' email + in-app
Admin action (any day)	Admin opens approval form	Admin fills new salary -> clicks Approve
Post-approval	System generates letter	New salary structure saved -> Increment Letter PDF generated -> Emailed to employee -> Reminder marked Actioned

13. Technology Stack (Revised)

Component	Technology	Purpose
Backend Framework	Django 4.2+ with DRF	Core API, views, auth, ORM
Database	PostgreSQL 15+	Primary data store; all relational data
API Authentication	DRF Token Auth + Session Auth	HR and Admin login; token for API consumers
Role-Based Access	Django Groups + custom permission mixin	Admin vs HR permission enforcement on every view/API

PDF Generation	WeasyPrint	All letters, slips, certificates from HTML/CSS templates
Excel Import / Export	django-import-export + openpyxl	Bulk data in/out for employees, students, salary
Email Engine	Django email (SMTP) / SendGrid	Exit secure links, increment notifications, certificate delivery
Background Tasks / Scheduler	Celery + Celery Beat + Redis	Daily anniversary check, bulk PDF jobs, async email sending
In-App Notifications	Django Channels + WebSocket OR polling	Real-time bell notifications; fallback to AJAX polling
File Storage	Django FileField + local /media/ (S3 optional)	Employee docs, generated PDFs, certificates
Secure Exit Links	UUID token in database with expiry timestamp	One-time or 7-day expiry; no login required for employee
Environment Config	python-dotenv (.env file)	All secrets, DB credentials, SMTP settings
Containerisation	Docker + docker-compose	PostgreSQL + Redis + Celery as services
Frontend	Django Templates + HTMX + vanilla JS / jQuery	Responsive HR portal UI; no separate React app in v1
Encryption (sensitive fields)	django-encrypted-fields or AES via cryptography lib	Aadhaar, PAN stored encrypted in DB
Audit Logging	Custom AuditLog model + Django signals	Track all create/update/delete by user + timestamp
Testing	pytest-django + factory_boy	Unit + integration tests for all critical flows

14. Database Model Summary

Model	Key Fields
User (Django built-in)	username, email, is_staff, is_superuser, groups
Department	id, name, created_at
Employee	id, emp_id (EMP-YYYY-XXXX), first_name, last_name, email, phone, dob, gender, address, department(FK), designation, employment_type, joining_date, notice_period_days, bond_period_months, aadhaar_encrypted, pan_encrypted, status (Active/Exited/Rejoined), created_by(FK User), created_at
EmergencyContact	id, employee(FK), name, relationship, phone
EmployeeDocument	id, employee(FK), doc_type, label, file, upload_date, uploaded_by(FK User)
SalaryStructure	id, employee(FK), effective_from, basic, hra, conveyance, medical, special, other_allowances(JSON), monthly_bonus, pf, professional_tax, tds, other_deductions(JSON), net_salary(calculated), created_by(FK User)

SalarySlip	id, employee(FK), payslip_no (PS-YYYY-MM-XXXX), month, year, working_days, days_worked, lop_days, one_time_bonus, one_time_deduction, gross, total_deductions, net_pay, payment_date, payment_mode, pdf_file, generated_by(FK User)
SalaryIncrementReminder	id, employee(FK), anniversary_date, reminder_15_sent, reminder_7_sent, reminder_0_sent, status (Pending/Actioned), actioned_by(FK User), actioned_at
SalaryIncrementApproval	id, employee(FK), reminder(FK), old_net, new_basic, new_hra, new_allowances, new_net, increment_amount, increment_pct, effective_date, reason, approved_by(FK User), approved_at, pdf_file
ExitRequest	id, employee(FK), resignation_date, last_working_day, exit_type, exit_reason, notice_waiver, notice_letter_required, status (Pending/InProgress/Completed), initiated_by(FK User), initiated_at
ExitSecureLink	id, exit_request(FK), token(UUID), created_at, expires_at, used
ExitFormResponse	id, exit_request(FK), asset_laptop, asset_id_card, asset_access_card, asset_others, kt_status, kt_handover_to, kt_remarks, reason_dropdown, reason_details, rating_env, rating_mgmt, recommend, personal_email, personal_phone, feedback, submitted_at
Student	id, name, email, phone, dob, institute, course_at_institute, student_type, program_name, department(FK), mentor, joining_date, completion_date, duration_display, project_description, cert_type, cert_no (CERT-YYYY-XXXXXX), status, cert_pdf, created_by(FK User)
Notification	id, recipient(FK User), notif_type, message, link, is_read, created_at
AuditLog	id, actor(FK User), action, model_name, object_id, description, timestamp

15. Non-Functional Requirements (Revised)

Category	Requirement
Performance	API < 2 sec (95th percentile). PDF single < 3 sec. Bulk PDF (50 slips) < 30 sec. Anniversary check task < 5 sec/day.
Security	HTTPS only in production. Session tokens expire after 8 hours of inactivity. Aadhaar/PAN encrypted at rest. Exit links UUID-based, 7-day expiry, single-use. File uploads: type + size validation (max 10 MB). Admin actions require re-confirmation for destructive operations.
Role Enforcement	Every API endpoint and view checks role via permission mixin. Admin-only endpoints return 403 to HR users.
Data Integrity	FK constraints enforced in PostgreSQL. Soft delete for employees (never hard delete). Salary history never deleted. All PDFs stored permanently.
Audit Trail	Every create, update, delete, and PDF generation logged in AuditLog with actor + timestamp.
Availability	99.5% uptime during business hours (9 AM – 6 PM IST). Monthly maintenance window.

Scalability	10,000 employees, 5,000 students, 5 years of salary slips without schema changes. Celery handles bulk async jobs.
Usability	Responsive for 1280px+ desktop screens. All forms have inline validation. Success/error toast messages on every action.
Compliance	Personal data (Aadhaar, PAN) access-controlled. No third-party data sharing. Email contains no sensitive data in plain text.

16. Revised Timeline

Phase	Activities	Duration
Setup & Auth	Project init, folder structure, PostgreSQL, Admin + HR roles, login/logout, permission middleware	2 days
Module 1 – Onboarding	Employee model, add/edit form, document upload, offer/appointment/bond letter PDF, employee list, export	5 days
Module 2 – Exit	Exit model, initiate form, secure link email, employee self-service form, status tracking, relieving/experience/NOC/F&F PDFs	6 days
Module 2B – Re-joining	Ex-employee search, re-onboard form, re-link history, certificate re-generation	2 days
Module 3 – Salary Slip	Salary structure form, monthly generation, LOP handling, bulk PDF ZIP, import/export	5 days
Increment Notification	Celery Beat task, SalaryIncrementReminder model, 15/7/0 day emails, approval form, increment letter PDF	4 days
Module 4 – Students	Student model, all certificate types, bulk PDF ZIP, import/export, email certificate	4 days
Notification System	Notification model, in-app bell, email engine wiring for all events	3 days
Dashboard	Admin dashboard (all widgets), HR dashboard, recent activities feed, pending actions widget	3 days
Audit Log & Security	AuditLog model, signals, Admin audit view, encrypted fields, security hardening	2 days
Testing	pytest unit tests for all critical flows, form validations, PDF generation, secure link expiry	3 days
Deployment & Docs	Docker setup, .env guide, user manual, Postman collection, deployment guide	3 days
Total		42 working days